Validated Low-Speed CFD of a Blended-Wing-Body Aircraft

Executive Summary

Kevin Armstrong II · UC Berkeley Mechanical Engineering · January 2026

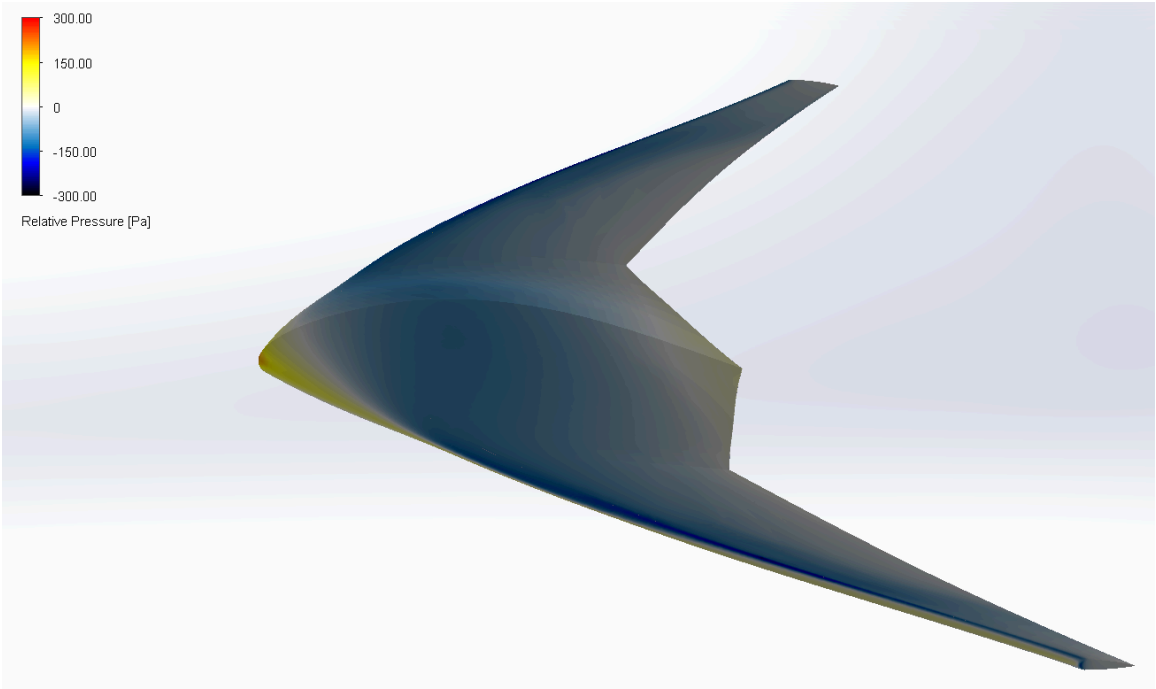

Figure ES-0a. BWB concept overview.

Objective

This project quantifies low-speed aerodynamics of a blended-wing-body (BWB) concept using steady-state CFD. Beyond reporting performance trends, the work establishes a repeatable analysis workflow (convergence, domain sensitivity, mesh sensitivity, consistent post-processing) suitable for early-stage concept screening. Outputs include CL, CD, L/D, and qualitative flow features from streamwise velocity cuts and surface pressure/Cp maps across an AoA sweep.

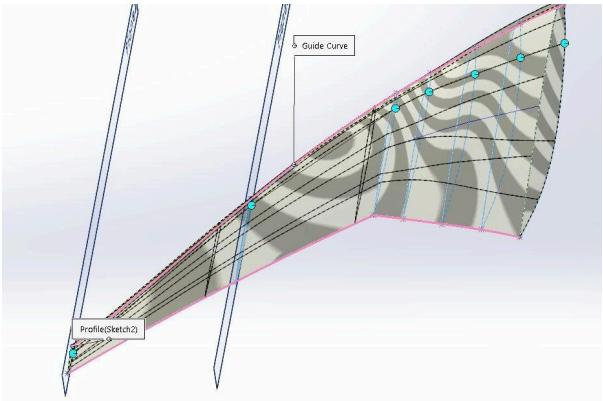

Figure CAD-1. SOLIDWORKS loft setup showing guide curves and multi-section profiles used to define the BWB outer mold line.

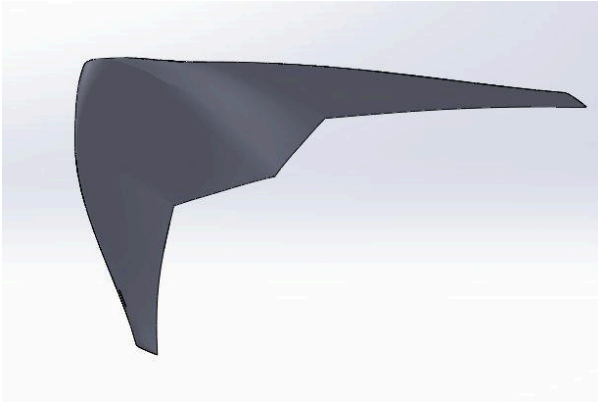

Figure CAD-2. Rendered SOLIDWORKS CAD model of the BWB lifting body.

Geometry & Reference Quantities

Configuration: Swept planform BWB lifting body modeled as an external aerodynamics case.

Approximate size: Span ≈ 8 in (0.203 m), length ≈ 9 in (0.229 m), peak thickness ≈ 2 in (0.051 m).

Reference area: The aerodynamic reference area S_{ref} was measured by sketching a top-view silhouette on the Top Plane and using SOLIDWORKS Measure. A force-to-coefficient consistency check confirms the effective S_{ref} used by the solver:

$$\mathbf{S_{ref} \approx 9.62 \times 10^{-3}\ m^2\ (\approx 14.9\ in^2)}$$

From span b ≈ 0.203 m, the implied mean chord is:

$$\mathbf{\bar{c} \approx S_{ref} / b \approx 0.047\ m\ (\approx 1.86\ in)}$$

Coordinate convention: +X is freestream direction; +Z is "up." Positive AoA implemented via VZ > 0, producing positive CL.

NACA-Derived Centerbody Profile (Mid-Chord)

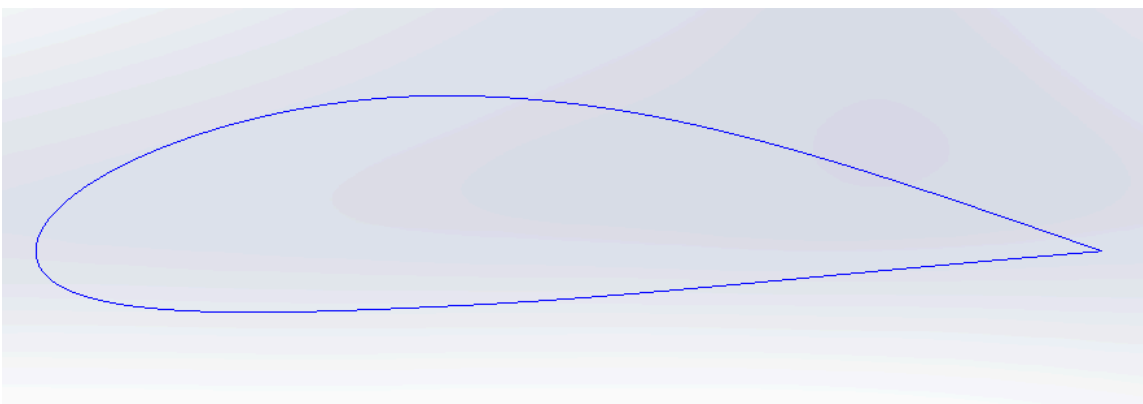

Figure ES-0. NACA-derived custom mid-chord centerbody section used in the BWB loft.

The section was generated from a NACA 00-series thickness distribution with a controlled centerbody volume term to support blending and internal packaging while maintaining a smooth leading edge and tapered trailing edge.

Freestream Conditions & Similarity Parameters

Working fluid: Air (SOLIDWORKS default).

Ambient conditions: p_∞ = 101,325 Pa, T_∞ = 293.2 K.

Freestream properties were measured upstream in an undisturbed region using Point Parameters rather than surface averages:

- $\rho_\infty \approx 1.20\ kg/m^3$
- $\mu_\infty \approx 1.8146 \times 10^{-5}$ Pa·s
- $V_\infty \approx 40$ mph = 17.88 m/s
- Dynamic pressure: $q_\infty = \frac{1}{2} \rho_\infty V_\infty^2 \approx 192$ Pa
- Reynolds number (mean chord): $Re = \rho_\infty V_\infty \bar{c} / \mu_\infty \approx 5.6 \times 10^4$

This is a low-Reynolds-number regime, where boundary layer behavior and separation sensitivity can strongly influence drag and maximum lift.

AoA Implementation

AoA was implemented by holding geometry fixed and changing the inlet velocity direction using a 3-component velocity vector. Example for the 4° case:

- VX = 39.9 mph, VY = 0 mph, VZ = 2.79 mph
- AoA α = arctan(VZ / VX) ≈ arctan(2.79/39.9) ≈ 4°

Physics Model & Boundary/Wall Inputs

Analysis type: External aerodynamics, steady-state.

Flow type: Laminar and Turbulent (turbulence modeling enabled).

Turbulence inputs: Intensity = 1%, length scale = 0.01 m.

Walls: Adiabatic, roughness = 0 μm.

Convergence Protocol

Each AoA case was run until goals were marked Achieved, then continued for an additional +100 iterations to confirm stability. Reported coefficients are based on the stabilized averaged goal values after this additional iteration window.

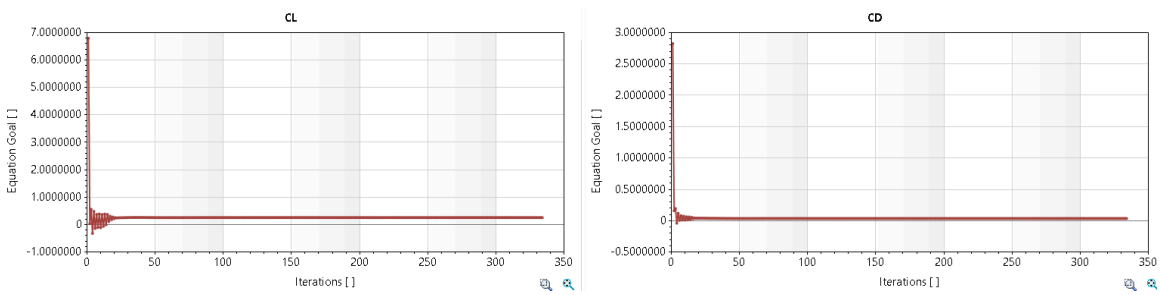

Figure ES-1. Convergence of aerodynamic goals at α = 4°. Lift and drag coefficient goals reach the solver's "Achieved" criterion and remain stable during an additional +100-iteration verification window. Final reported coefficients use stabilized averaged values after this window to reduce sensitivity to transient iteration-to-iteration fluctuations.

Verification: Domain Independence (4° baseline, medium mesh)

Two computational domains were compared (all other settings unchanged). Coefficients show sub-1% sensitivity, supporting that far-field boundaries are not materially biasing the solution.

- Large domain: CL = 0.259792, CD = 0.0372691, L/D = 6.97073
- Small domain: CL = 0.260230, CD = 0.0374288, L/D = 6.95267

Percent differences (relative to large domain): ΔCL ≈ 0.17%, ΔCD ≈ 0.43%, Δ(L/D) ≈ 0.26%

Table ES-2. Domain sensitivity (α = 4° baseline; averaged values after +100-iteration verification)

Case	CL	CD	L/D	ΔCL (%)	ΔCD (%)	Δ(L/D) (%)
Large domain	0.259792	0.037269	6.970730	+0.00	+0.00	+0.00
Small domain	0.260230	0.037429	6.952670	+0.17	+0.43	-0.26

Figure ES-2. Domain sensitivity check at α = 4° (medium mesh). Aerodynamic coefficients change by <0.5% between two domain sizes, indicating the chosen far-field extents are sufficient for concept-level coefficient prediction.

Verification: Mesh Independence (4° baseline, large domain)

Three mesh densities were compared. Medium → fine changes are ~1% for CL and <0.5% for CD, indicating near-converged coefficient values for the sweep.

- Coarse: CL = 0.254886, CD = 0.0377031, L/D = 6.76034
- Medium (baseline): CL = 0.259792, CD = 0.0372691, L/D = 6.97073
- Fine: CL = 0.263079, CD = 0.0371315, L/D = 7.08507

Medium → fine deltas: ΔCL ≈ 1.25%, ΔCD ≈ 0.37%, Δ(L/D) ≈ 1.6%

Table ES-3. Mesh sensitivity (α = 4°, large domain; averaged values after +100-iteration verification)

Mesh	CL	CD	L/D
Coarse	0.254886	0.037703	6.76034
Medium (baseline)	0.259792	0.037269	6.97073
Fine	0.263079	0.037131	7.08507
Medium → Fine Δ (%)	+1.27%	-0.37%	+1.64%

Figure ES-3. Mesh sensitivity check at α = 4° (large domain). Medium → fine coefficient changes are small (≈1% for CL, <0.5% for CD), supporting use of the medium mesh for the AoA sweep with a fine-mesh verification point.

AoA Sweep Results (−2° to +8°)

Using the validated baseline setup (large domain + medium mesh) and consistent post-processing, converged coefficients were obtained for α = −2°, 0°, 2°, 4°, 6°, 8°:

AoA (°)	CL	CD	L/D
−2	−0.041385	0.028280	−1.46342
0	0.058617	0.027234	2.15237
2	0.161187	0.030716	5.24760
4	0.259792	0.037269	6.97073
6	0.355393	0.048587	7.31458
8	0.447335	0.064402	6.94595

Blended Wing Body CFD Summary: Aerodynamic Coefficients vs Angle of Attack
Tabulated results (values used for plots)

AoA	CL	CD	LD
-2	-0.041385	0.028280	-1.46342
0	0.058617	0.027234	2.15237
2	0.161187	0.030716	5.24760
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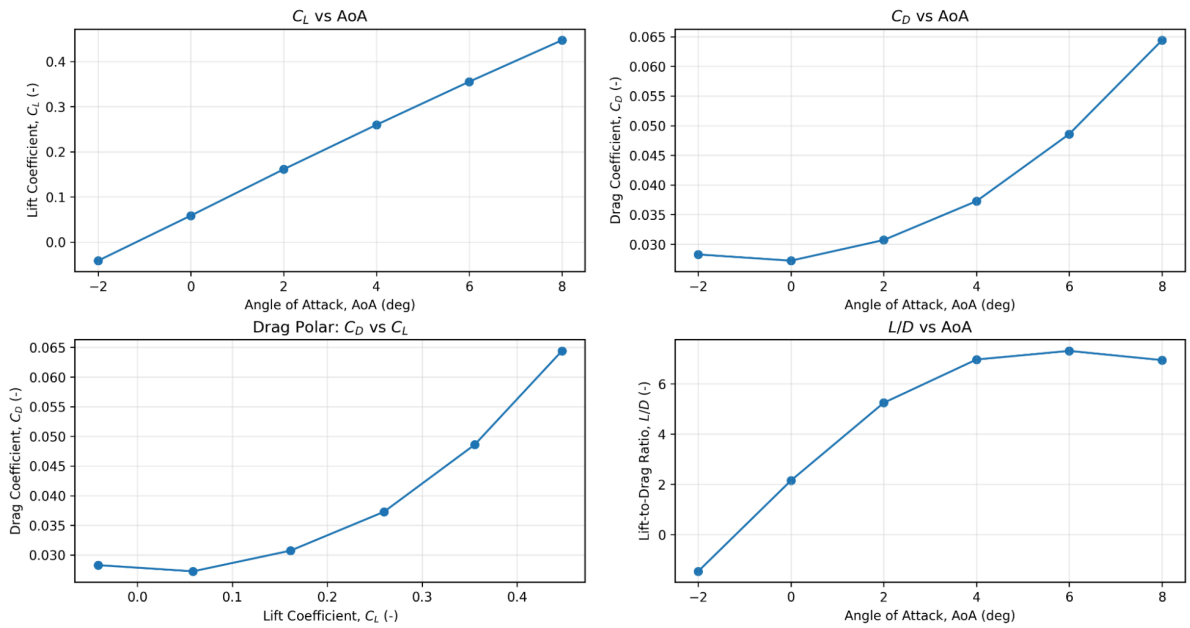

Figure ES-4. AoA sweep results for the BWB concept at V∞ ≈ 40 mph. CL increases nearly linearly through 6°, while CD increases with AoA after a low point near 0°. L/D peaks near 6° at ≈7.31 and declines by 8° as drag rises more rapidly.

Observed trends:

- Lift increases approximately linearly from 0° through 6°, consistent with pre-stall behavior.
- Drag is minimized near 0° and increases with AoA due to increased induced drag and growing wake losses.
- Peak aerodynamic efficiency (L/D) occurs near 6°, after which drag growth outpaces lift growth.

Flow Visualization Standard

For a single consistent velocity field comparison, streamwise velocity (VX) was selected instead of velocity magnitude |V|. VX most directly highlights streamwise acceleration over the body and wake momentum deficit downstream—features tied to aerodynamic loading and drag. Velocity magnitude mixes in crossflow (VZ, VY) created by AoA, which can obscure wake-loss comparisons across AoA cases.

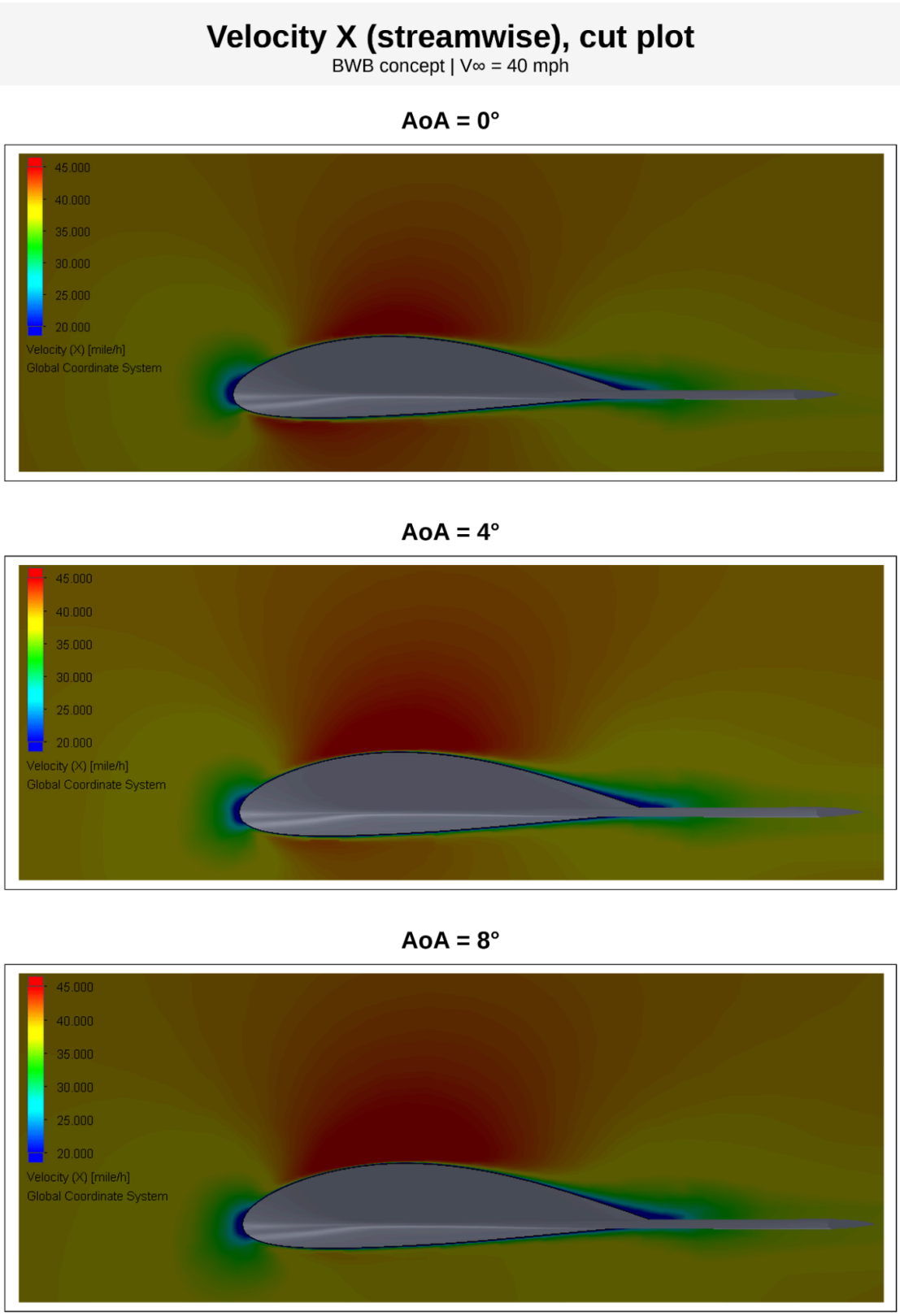

Figure ES-5. Streamwise velocity (VX) centerline cut plots across AoA. At α = 0°, 4°, and 8° (V∞ ≈ 40 mph), the near-body acceleration region strengthens with AoA and the downstream wake thickens, indicating increased circulation and momentum deficit consistent with rising aerodynamic loading and drag.

Pressure Coefficient Scaling

Surface maps were exported as relative static pressure, $\Delta p = p - p_\infty$, with a fixed legend scale of −300 to +300 Pa for direct comparison across cases. The pressure coefficient was computed using the measured freestream properties:

$$\mathbf{C_p = \Delta p / q_\infty = \Delta p / 192\ Pa \quad \Longrightarrow \quad C_p\ range \approx -1.56\ to\ +1.56}$$

As α increases, the suction region (negative Δp / Cp) strengthens over the forward/outer planform, consistent with the observed increase in CL in the pre-stall range.

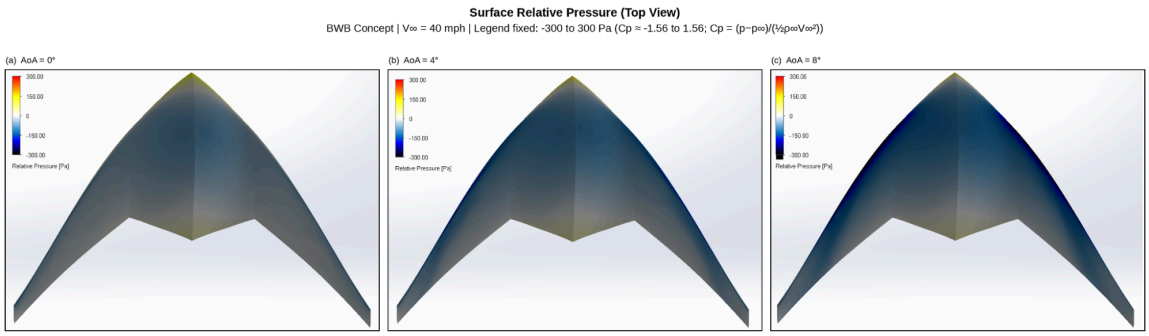

Figure ES-6. Relative static pressure, $\Delta p = p - p_\infty$, at (a) α = 0°, (b) α = 4°, and (c) α = 8° at $V_\infty \approx 40$ mph using a fixed color scale (−300 to +300 Pa). Using freestream values sampled upstream ($\rho_\infty \approx 1.20$ kg/m³, $q_\infty \approx 192$ Pa), this legend corresponds to $C_p \approx \Delta p/q_\infty \approx -1.56$ to +1.56.

Key Takeaways

- Quantified aerodynamic performance: The BWB exhibits increasing lift with AoA and achieves maximum L/D near 6° under tested low-speed, low-Re conditions.
- Verification and credibility: Domain and mesh sensitivity studies show reported coefficients are numerically stable (sub-1% domain sensitivity; ~1% medium→fine sensitivity for CL).
- Consistent visualization standards: Identical camera views, fixed legends, and VX-based slices support clean qualitative comparisons across AoA.
- Clear nondimensional scaling: Relative pressure fields are tied to Cp using measured far-field properties, enabling standard aerodynamic interpretation.

Planned Wind Tunnel Validation & Tufting Test

Physical wind tunnel testing is planned to validate the CFD predictions. The 3D-printed BWB model has been prepared for tufting—a qualitative flow visualization technique that reveals surface separation and reattachment patterns across the planform. Key pre-test calculations are summarized below.

Reynolds Number & Dynamic Similarity

The CFD baseline was run at V_∞ = 40 mph (17.88 m/s) over mean chord $\bar{c}$ = 0.047 m, giving Re ≈ 5.56×10^4. Since the 3D-printed model is 1:1 scale, running the wind tunnel at the same 40 mph matches the CFD Reynolds number exactly—no scale correction needed.

Expected Aerodynamic Forces (by Tunnel Speed)

Predicted lift and drag forces at CL ≈ 0.26 (4° AoA) and CD ≈ 0.037:

V (m/s)	V (mph)	Re	q_∞ (Pa)	Lift (N)	Drag (N)
10.0	22.4	3.11×10^4	60.0	0.150	0.021
15.0	33.6	4.66×10^4	135.0	0.338	0.048
17.9	40.0	5.56×10^4	191.8	0.480	0.068
20.0	44.7	6.22×10^4	240.0	0.600	0.085
25.0	55.9	7.77×10^4	375.0	0.938	0.134
30.0	67.1	9.32×10^4	540.0	1.351	0.192

Forces scale as V^2. At 40 mph (Re-matched), expected lift ≈ 0.48 N and drag ≈ 0.068 N at 4° AoA—small forces requiring a sensitive load cell for quantitative measurement.

Tunnel Blockage

Blockage ratio = model frontal area / tunnel cross-section area. Estimated model frontal area (span × peak thickness at 0° AoA) ≈ 103.5 cm². Target <10%.

Tunnel Size	Tunnel Area (cm²)	Blockage	Assessment
12 × 12 in	929	11.1%	⚠ Marginally high—use larger tunnel or apply correction
18 × 18 in	2,090	5.0%	✓ Acceptable
24 × 24 in	3,716	2.8%	✓ Good

Tufting Guidelines

Tufts are short, lightweight strands of yarn attached to the model surface to visualize local flow direction. They deflect, oscillate, or reverse when separation occurs, giving a qualitative map of attached vs. separated regions across AoA.

Parameter	Recommendation	Basis
Tuft length	~7 mm (0.28 in)	~15% mean chord
Spanwise spacing	~5.6 mm (0.22 in)	~12% mean chord
Chordwise rows	3–5 rows	LE, mid, aft, near TE
Est. total tufts	~36 per chordwise row	Full span / spacing
Material	Lightweight sewing thread or wool	Low mass, visible in video

- Attach tufts with small drops of tape or glue—ensure they are free to move and not kinked.
- Record with high-frame-rate video at each AoA for comparison with CFD surface Cp maps.
- Key regions: leading edge suction peak, outer panel, wing-body blend junction, and trailing edge.
- Tuft behavior directly compared to CFD Cp maps to assess whether predicted separation patterns match physical flow.

Validation Objectives

- Confirm qualitative agreement between tuft separation patterns and CFD surface Cp maps across AoA sweep.
- Identify discrepancies in separation onset AoA between CFD and physical model (important at low Re ≈ 5.6×10^4).
- Provide qualitative stall AoA estimate for comparison with CFD L/D peak near 6°.
- [Quantitative forces and moments pending load cell availability.]

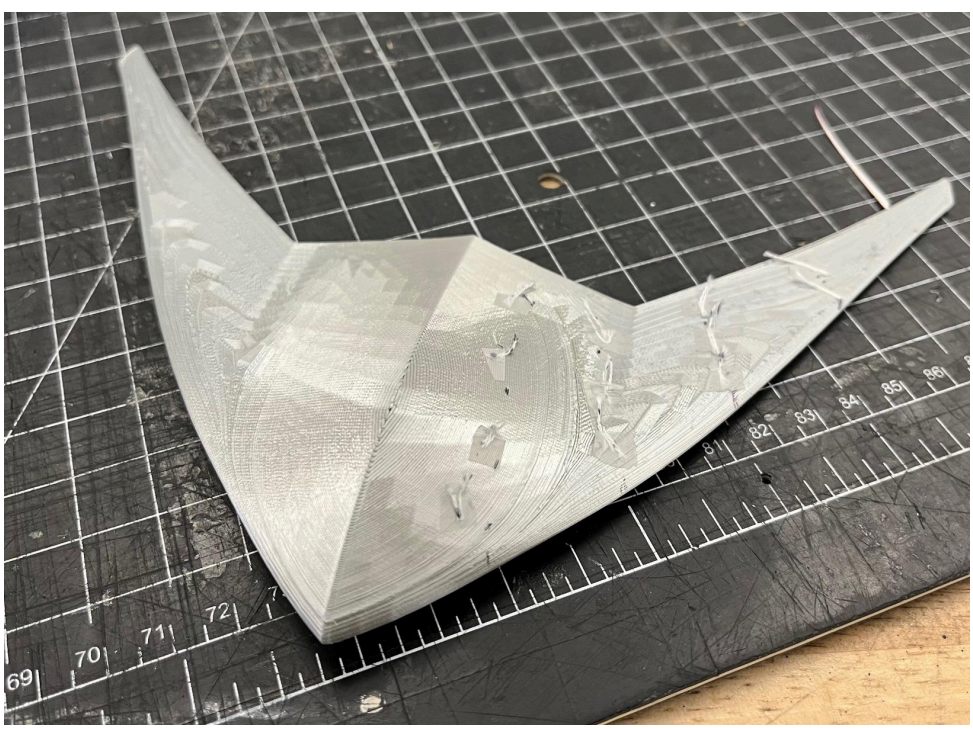

Figure WT-1. 3D-printed BWB model prepared for initial tufting—tufts will be applied prior to wind tunnel testing to visualize surface flow patterns and validate CFD predictions.

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- Figure WT-1: 3D-printed model prepared for tufting (wind tunnel validation)

Wind tunnel test results to be appended upon completion.